PhysEval: Quantifying the Gap Between Video Generation and World Physical Laws

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Abstract

Text-to-video (T2V) models can now synthesize visually compelling motion, but visual plausibility does not guarantee that the generated world follows the physical laws it depicts. A falling object, sliding block, collision, or oscillating spring may look reasonable while still implying an incorrect acceleration, material parameter, or conservation relationship. Most existing benchmarks emphasize perceptual quality, text-video alignment, or qualitative physical plausibility, leaving a gap between visually plausible video generation and quantitatively verifiable physical behavior. To quantify this gap, we present PhysEval, a benchmark dataset and automatic evaluation protocol for scoring the physical accuracy of T2V generations. PhysEval pairs 500 prompts with auditable metadata specifying the structured scene configuration, target physical values, and calibration references when applicable. It covers ten physical metrics across kinematics, physical constants, material parameters, and conservation laws. Given generated videos, PhysEval automatically determines which samples are measurable, estimates the relevant physical quantities, and produces normalized scores together with discard diagnostics. Our evaluation of 3,000 videos shows that current T2V models still struggle to reliably generate videos that are both measurable and physically consistent: even visually plausible samples often fail to express the quantitative cues needed by the underlying physical law. Code and benchmark are available at: https://github.com/Cryptic0429/PhysEval.

1 Introduction

Text-to-video (T2V) generation has evolved from early GAN-based and autoregressive approaches to large-scale diffusion and transformer-based foundation models, enabling the generation of increasingly detailed scenes, temporally coherent motions, and semantically aligned videos conditioned on textual prompts [1–7]. Beyond creative media generation, these models are increasingly being explored as potential world models capable of modeling complex visual dynamics and supporting applications in simulation and embodied intelligence. These applications require a stronger notion of correctness than visual realism. A generated falling object may look natural while accelerating at the wrong rate, a sliding block may imply an inconsistent friction coefficient, a collision may create or lose momentum, and an oscillating spring may have a period that does not match the specified mass and stiffness. In such cases, the video may satisfy human perceptual expectations while violating the underlying physical principles of the depicted scene.

Existing evaluations only partially address this issue. Classical metrics such as IS [8], FID [9], FVD [10], and CLIP-based similarity [11, 12] provide aggregate signals about sample quality, distributional realism, or text-video alignment. Recent benchmark suites go further by decomposing video generation into perceptual, temporal, semantic, and compositional dimensions [13–16]. Physics-aware benchmarks have begun to assess whether generated videos are consistent with physical

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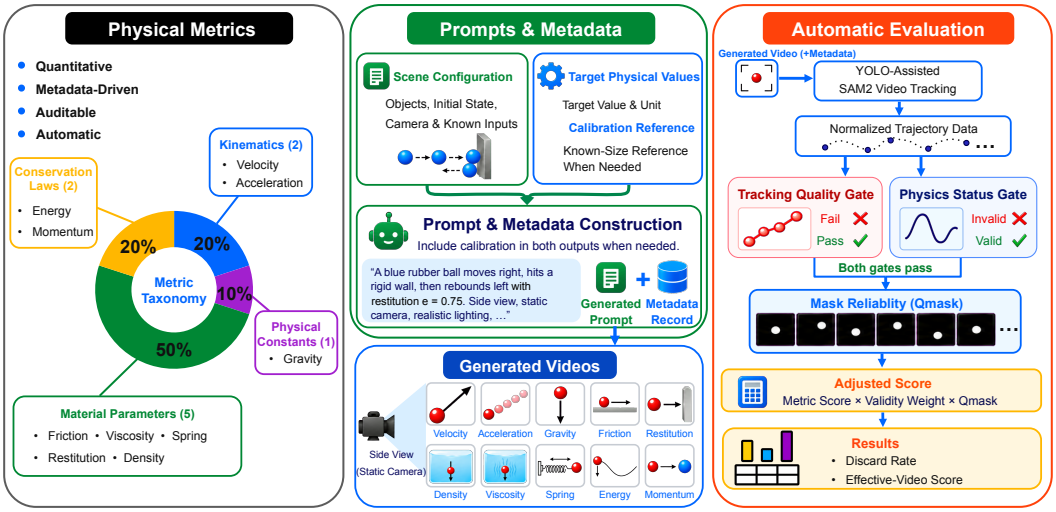


Figure 1: Overview of PhysEval. PhysEval organizes quantitative physical evaluation into physical metrics, prompts and metadata construction, and automatic evaluation. Each metric is paired with prompts and auditable metadata, generated videos are sampled from T2V models, and YOLO-assisted SAM2 tracking with task-specific physics evaluators produces discard diagnostics and effective-video scores for measuring physical accuracy.

common sense and fundamental physical principles [17–21]. However, most of these evaluations still ask whether a phenomenon looks plausible, whether a generated event follows a coarse physical rule, or whether an evaluator can classify a physical outcome. They do not directly answer a more diagnostic question: when a prompt specifies a physical quantity, does the generated video contain measurable motion whose estimated value matches the requested target? Answering it requires more than a perceptual score or a binary physical-plausibility judgment. It requires prompts tied to explicit numerical targets, trajectory recovery from generated videos, task-specific inverse physical estimators, and diagnostics that distinguish inaccurate physics from unmeasurable generations.

To address this gap, we introduce PhysEval, a benchmark dataset and automatic protocol for quantitative physical evaluation of T2V generations. Specifically, PhysEval pairs 500 prompts with auditable metadata: prompts define the generation requirements for T2V models, while metadata specifies the structured scene configuration, target physical values, and calibration references. We first initialize objects and track masks to recover visible trajectories; we then apply the hard Tracking Quality Gate and Physics Status Gate to determine eligibility, and weight the physical-accuracy score by a soft mask-reliability score. Each sample records its prompt, target, measured value, relative error, tracking-quality status, physics status, mask-reliability score, and discard diagnostics.

Based on PhysEval, we evaluate current open-source and closed-source T2V models under this protocol and find that quantitative physical generation remains highly unreliable. The results show two distinct failure modes. First, many generated videos cannot be measured because the object is static, the expected object count is wrong, tracking is unstable, or the physical event is not valid under the evaluator assumptions. Second, even among videos that pass these gates, model performance remains weak and highly metric-dependent. Performance on viscosity, conservation laws, and several motion-based measurements remains poor. For this reason, our scoring separates effective-video scores from discard diagnostics. Effective-video scores measure physical accuracy only on measurable videos, while discard rates reveal how often a model fails to produce a sample that can support the requested physical measurement. Together, these findings show that current T2V models can often generate visually plausible physical scenes, but still struggle to achieve reliable quantitative control over physical behavior.

In summary, our main contributions are:

- We introduce PhysEval, a quantitative benchmark for evaluating whether T2V generations match specified physical quantities across kinematics, physical constants, material parameters, and conservation laws.

- We develop an auditable evaluation pipeline that links prompts to structured physical metadata, tracks objects in generated videos, estimates task-specific physical quantities, and reports tracking quality, physics status, mask reliability, and discard diagnostics.
- We benchmark representative T2V models and reveal substantial limitations in quantitative physical accuracy: many generated videos are unmeasurable, while measurable videos exhibit pronounced metric-dependent weaknesses.

2 Related Work

2.1 Video Generation

Modern text-to-video (T2V) systems differ mainly in how they represent temporal dynamics. One line extends diffusion models from images to videos, adding temporal modules, latent video spaces, cascaded refinement, or controllable motion interfaces [1–3, 22–31]. Representative systems such as Make-A-Video [2], Imagen Video [3], LVDM [25], and VideoCrafter1 [28] show how image priors, latent denoising, and video super-resolution can be combined to improve visual quality and temporal coherence. Another line replaces U-Net denoisers with transformer-heavy diffusion architectures, improving scale and long-range spatiotemporal modeling in systems such as Latte [32], CogVideoX-5B [4], HunyuanVideo-1.5 [5, 6], Wan2.1-1.3B [7, 33], Wan2.2-A14B [34], Open-Sora Plan [35], and Movie Gen [36]. A third line formulates video synthesis as autoregressive or language-model-style prediction over visual or multimodal tokens; examples include CogVideo [37], NUWA [38], Phenaki [39], VideoPoet [40], and MAGI-1 [41]. These model families have pushed fidelity, prompt following, and duration, but they do not by themselves ensure that generated trajectories satisfy measurable physical laws.

2.2 Evaluation of Video Generation

Metric-based evaluation. Automatic evaluation often relies on distributional, embedding, or learned perceptual metrics. Inception Score [8] and FID [9] measure image-level realism and distributional similarity; FVD [10] uses video features learned from action-recognition data [42, 43]; and CLIP [11] and CLIPScore [12] estimate text-visual alignment with vision-language embeddings and captioned-video data [44]. Video quality assessment methods model perceptual quality, temporal artifacts, and aesthetics [45–49], and some pipelines reuse optical flow [50] or self-supervised features [51] for motion and temporal consistency. These metrics are efficient, but they can miss localized causal errors and numerical physical inconsistencies.

Human-subjective evaluation. Human studies, mean opinion scores (MOS), and preference annotations evaluate visual realism, temporal naturalness, and human-aligned quality, often training learned evaluators from subjective labels. Human-subjective evaluation typically relies on human studies, mean opinion scores (MOS), and pairwise preference annotations to assess multiple aspects of generated video quality, such as visual fidelity, temporal consistency, and alignment with input prompt. To facilitate the development of reliable and subjective-aligned video quality assessment (VQA) methods, recent efforts have increasingly focused on constructing large-scale annotated datasets and developing learned evaluators for diverse AI-generated and multimedia content [52–57]. Such protocols capture perceptual artifacts that generic automatic metrics overlook, but they are costly, annotator-dependent, and do not recover explicit physical quantities from object trajectories.

Benchmark-based evaluation. Benchmark suites provide more structured evaluation than single metrics or isolated user studies. General-purpose benchmarks such as VBench [13], EvalCrafter [14], FETV [15], and T2V-CompBench [16] cover perceptual quality, temporal consistency, text-video alignment, fine-grained prompt categories, and compositional skills. Their targets, however, remain mostly perceptual, semantic, subjective, or compositional. Physics-oriented benchmarks move closer to world simulation: VideoPhy evaluates physical plausibility in activity prompts [17], PhyGenBench uses physical-law prompts with a hierarchical VLM/LLM evaluator [18], Physics-IQ probes physical principles [19], Morpheus scores real experiments with conservation and equation-fitting signals [20], and WorldBench [21] and 4DWorldBench [58] add physical-parameter and 3D/4D diagnostics. PhysEval differs by targeting T2V outputs directly (Table 1): each prompt is tied to explicit numerical

Table 1: Comparison with representative physics-oriented benchmarks. T2V: text-to-video; I2V: image-to-video; V2V: video-to-video; T/I/V: text/image/video. "Automatic" denotes evaluation without per-video human labels at evaluation time; "Quantitative" denotes explicit physical quantities or physical-quantity scores.

Benchmark	Generation task	Evaluation focus	Automatic	Quantitative
VideoPhy [17]	T2V	Physical commonsense plausibility	partial	–
PhyGenBench [18]	T2V	Qualitative physical-law compliance	✓	–
Physics-IQ [19]	I2V / V2V	Future-frame spatiotemporal alignment	✓	partial
Morpheus [20]	Conditioned I2V	Dynamical invariants & conservation	partial	✓
WorldBench [21]	Video prediction / I2V	Intuitive physics & parameter diagnostics	✓	partial
4DWorldBench [58]	T/I-to-3D & T/I/V-to-4D	4D geometric & physical realism	✓	–
PhysEval (Ours)	**T2V**	**Quantitative physical-parameter adherence**	✓	✓

Table 2: Task taxonomy. Every metric has 50 prompts. *Obj.* denotes the number of tracked target objects, and *Min. records* are estimator prerequisites rather than guarantees of valid measurement. Momentum additionally requires at least 12 common frames for event windows.

Category	Metric	Obj.	Min. records	Estimate / additional information
Kinematics	Velocity	1	4	Linear fit; spatial scale
Kinematics	Acceleration	1	5	Quadratic fit; spatial scale
Constant	Gravity	1	5	Vertical quadratic fit; spatial scale
Material	Friction	1	8	Deceleration / known gravity
Material	Restitution	1	10	Pre/post-impact speed ratio
Material	Density	1	8	Early acceleration; fluid density, scale
Material	Viscosity	1	12	Terminal speed; radius, densities, scale
Material	Spring	1	12	Period; known mass and time base
Conservation	Energy	1	10	Drop / kinetic-energy ratio; scale, gravity
Conservation	Momentum	2	8 each	Two tracks, masses, aligned event windows

physical targets, and those targets are estimated from generated videos through object tracking and task-specific inverse physics.

3 Benchmark Datasets

3.1 Physical Task Taxonomy

The benchmark comprises 500 prompt–metadata pairs, evenly distributed across ten physical metrics (50 prompts each). As shown in Figure 1, these metrics span four core categories: kinematics (2 metrics: velocity and acceleration), physical constants (1: gravity), material parameters (5: friction, restitution, density, viscosity, and spring constant), and conservation laws (2: mechanical energy and 1D momentum), accounting for 20%, 10%, 50%, and 20% of the benchmark, respectively. Gravity is treated as a separate physical constant because the evaluator estimates scene-level gravitational acceleration rather than generic object acceleration. These categories are selected because they can be estimated from short monocular videos under idealized physical models and controlled planar-motion assumptions, while still requiring nontrivial object motion and temporal consistency.

Each prompt is assigned a single metric and numerical target, while object descriptions and scene templates are systematically varied under a unified fixed-camera, approximately planar viewing setup. Rather than drawing from a continuous uniform distribution, target values are defined on a discrete, auditable grid to support reproducible evaluation. Specifically, velocity spans ten target values with unequal prompt frequencies, and acceleration covers thirteen distinct targets, including negative values for deceleration. Gravitational acceleration is fixed at 9.8 m/s^2. Density is evaluated across three levels (850, $1{,}100$, and $1{,}400$ kg/m^3, with 16, 17, and 17 prompts, respectively). Mechanical energy conservation uses a target ratio of 1, and momentum conservation specifies a relative-error target of 0 rather than zero absolute momentum. Appendix A details the complete set of target values and their frequencies, while Appendix B documents the inverse modeling assumptions.

3.2 Dataset Specification and Access

Figure 2 conceptually represents prompt composition as $P_i = T_{m_i}(S_i, R_i, K_i)$, where metric type m_i selects template T_{m_i}, S_i specifies scene configuration (objects, initial state, camera, and known

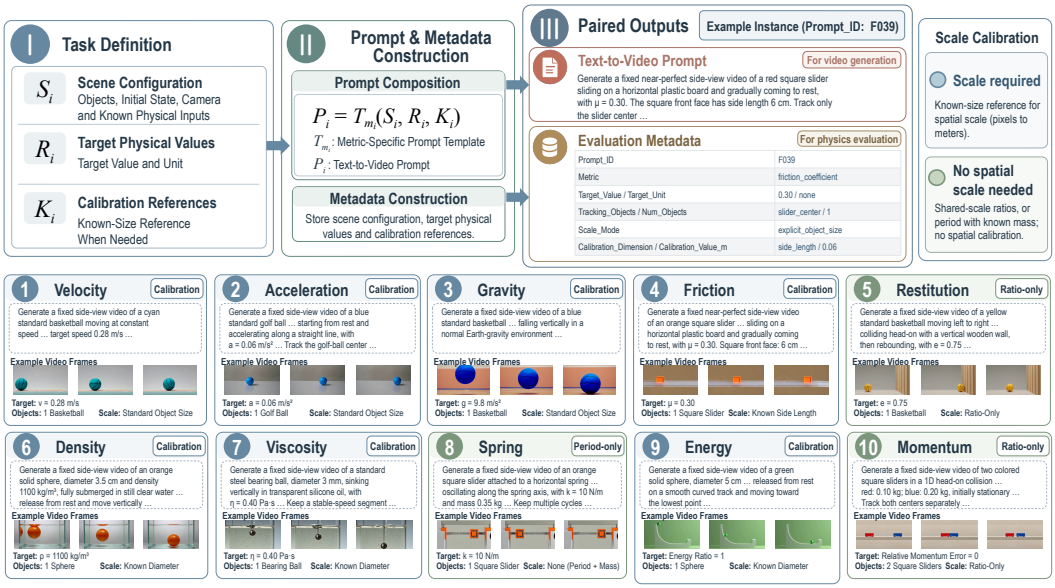


Figure 2: Overview of prompt–metadata construction and ten task examples. The upper panel illustrates how scene configuration (S_i), target physical values (R_i), and calibration references (K_i) are compiled into paired prompts and evaluation metadata; the lower panel presents one example per metric with its prompt excerpt, target value, calibration mode, and keyframes from generated videos (see Appendix B for estimator details).

inputs), R_i specifies target physical values (target value and unit), and K_i provides calibration references when needed (empty otherwise). LLMs interactively expand these inputs into fluent prompts while keeping visual styling solely in P_i; targets in R_i appear either as explicit numerals or implicitly via context (e.g., Earth gravity in P_i stores $g = 9.8$ m/s^2 for scoring). Each prompt and metadata record share a `Prompt_ID`, and m_i fixes the physics estimator and measurement method.

Estimating velocity, acceleration, gravity, friction, density, viscosity, and energy requires spatial calibration to convert pixels into meters. When spatial scale is needed, prompt P_i specifies either a standard-sized object (e.g., a basketball) or explicit dimensions (e.g., a 6 cm slider), while metadata K_i records the reference object, dimension type, and metric length. Image trajectories are scaled under the assumption of a fixed camera and a motion plane approximately parallel to the image plane. A shared spatial scale cancels for restitution and relative momentum error, while spring stiffness uses oscillation period, time base, and known mass (Appendix B.1).

4 Evaluation

To separate unmeasurable generations from inaccurate physics, our evaluation protocol filters generated videos into an *effective set* defined by two hard validity gates: the Tracking Quality Gate and the Physics Status Gate. A video is effective only when the tracker recovers the expected measurable motion and the task-specific evaluator returns `valid` or `weak_valid`. Although the implementation checks tracking first to avoid unnecessary computation on untrackable videos, conceptually these gates are parallel requirements for measurability, so failed videos are diagnosed as failing tracking only, failing physics only, or failing both. Only after both gates pass does the scorer compute the mask-reliability weight, relative error, physical-accuracy score, and adjusted final score.

4.1 Ordered Validity Gates

Within the YOLO family [59], YOLOv8n [60] provides initial proposals, while motion-based initialization handles insufficient detections. SAM2 [61] propagates selected masks across frames to recover object trajectories.

Tracking Quality Gate. The hard tracking policy checks persistent object count, $\geq 50\%$ frame coverage, and primary-object motion of ≥ 0.75 object diameters. Corresponding to Figure 1, this

gate defines the binary tracking-validity indicator M_{track} as:

$$M_{\text{track}} = \begin{cases} 1, & \text{tracking passes,} \\ 0, & \text{tracking fails under the hard policy.} \end{cases} \tag{1}$$

Physics Status Gate. Task-specific estimators return `valid`, `weak_valid`, `invalid`, or `failed`. Trajectory-fitting tasks (velocity, acceleration, gravity, friction, and density) share two R^2 thresholds ($R^2 \geq 0.85$: `valid`; $0.65 \leq R^2 < 0.85$: `weak_valid`; $R^2 < 0.65$: `invalid`; Appendices B–C), defining the physics-validity weight M_{valid} as:

$$M_{\text{valid}} = \begin{cases} 1, & \texttt{valid}, \\ 0.8, & \texttt{weak_valid}, \\ 0, & \texttt{invalid} \text{ or } \texttt{failed}. \end{cases} \tag{2}$$

Mask Reliability (Q_{mask}). Mask stability over the middle 15–85% window uses ten diagnostics, including centroid step with good/bad thresholds of $2.50/5.00$ object diameters (Appendix C.3). For each tracked object o with available diagnostic set $\mathcal{A}_o \neq \emptyset$ and each diagnostic $j \in \mathcal{A}_o$, normalized degradation $p_{oj} \in [0, 1]$ maps to $q_{oj} = 1 - p_{oj}^{0.35}$. The object score u_o, video-level reliability Q_{mask}, and measurement multiplier M_{measure} are defined as:

$$u_o = \min_{j \in \mathcal{A}_o} q_{oj}, \qquad Q_{\text{mask}} = \min_o u_o, \qquad M_{\text{measure}} = M_{\text{track}} Q_{\text{mask}}. \tag{3}$$

Missing components are ignored, whereas insufficient valid middle frames or a low-coverage flag without a measured coverage component assigns $u_o = 0$; every scored object must retain at least one computable component, and quality-control flags do not alter component scores. Exact zero scores ($Q_{\text{mask}} = 0$) remain zero without removing effective eligibility.

4.2 Object Tracking and Component Selection

We compared YOLOv8n+SAM2 with Faster R-CNN [62]+SAM2 and YOLOv8n paired with the Track Anything Model (TAM) [63] on the same 3,000 videos from six generators. They achieved run-completion rates of 93.47%, 93.77%, and 93.40%, respectively, where run completion denotes successful execution with saved tracking outputs prior to post-run checks on object-count consistency, frame coverage, or motion extent. These similarly high rates confirm the operational stability of the tracking workflow across the tested combinations, with YOLOv8n+SAM2 used as the default.

4.3 Scoring Formula

Let (u_i, v_i) be the tracked object center at timestamp t_i in frame i. Trajectory fitting and physics-validity checks are first conducted in the pixel domain before metric conversion, ensuring that validity decisions do not depend on the physical scale. For calibrated tasks, with reference length L in K_i and object pixel size ℓ—measured across valid frames as the median equivalent mask diameter (diameter mode) or mean median box width and height (side-length mode)—the spatial scale α (in meters/pixel) is defined as:

$$\alpha = L/\ell. \tag{4}$$

With a reference pixel origin (u_0, v_0), the 2D metric position vector $\mathbf{r}_i$ is defined as:

$$\mathbf{r}_i = \alpha \begin{bmatrix} u_i - u_0 \\ v_0 - v_i \end{bmatrix}, \tag{5}$$

whose planar coordinates support 2D kinematic analysis (e.g., computing velocity components v_x, v_y in energy conservation). For tasks governed by a single dominant axis (e.g., horizontal sliding, vertical free fall), projecting (u_i, v_i) onto the motion axis yields a 1D pixel coordinate q_i, which isolates the primary trajectory from orthogonal tracking jitter. The corresponding 1D metric coordinate $s_i = \alpha q_i$ then converts displacements into meters, allowing linear slope ($\hat{v}$) or quadratic acceleration ($\hat{a} = 2\alpha\hat{a}_{2,\text{px}}$) to be fitted directly in metric units (scaling preserves fit R^2, while ratio/period tasks require no spatial scale; Appendix B). For target y and estimate $\hat{y}$, the relative error RelErr for nonzero targets is defined as:

$$\text{RelErr} = \frac{|\hat{y} - y|}{\max(|y|, 10^{-12})}, \tag{6}$$

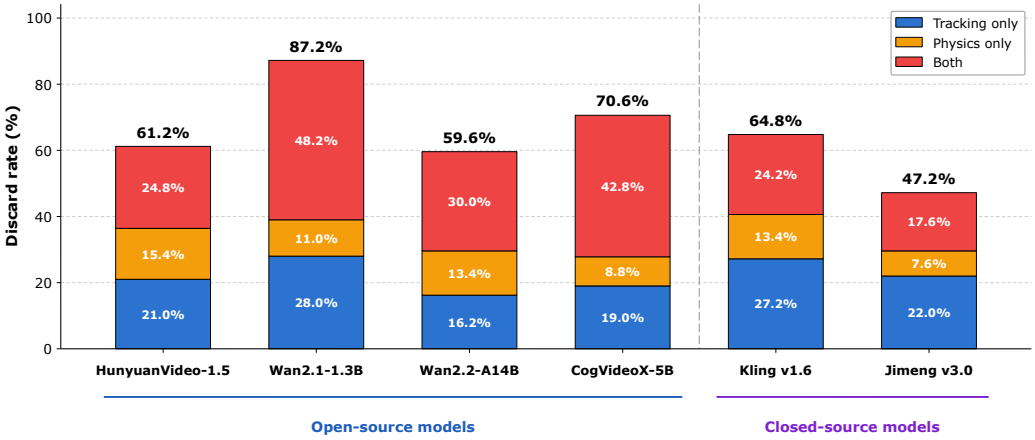


Figure 3: Discard outcomes over 500 videos per model, grouped by open-source and closed-source models. Bars are stacked by diagnostic source: tracking-only failures, physics-only failures, and failures of both hard gates.

where 10^{-12} avoids division by zero (for momentum, $\hat{y}$ is already a relative error with $y = 0$, so $\mathrm{RelErr} = |\hat{y}|$). For tolerance τ_k (0.30 for velocity, gravity, restitution; 0.40 for viscosity; 0.35 otherwise), the base physical-accuracy score $S_{\mathrm{phys}} \in [0, 100]$ is defined as:

$$S_{\mathrm{phys}} = 100 \exp(-\mathrm{RelErr}/\tau_k). \tag{7}$$

With multipliers M_{valid} and M_{measure} and N_{eff} effective videos out of N_{total}, the adjusted per-video score S_{video} and discard rate R_{discard} are defined as:

$$S_{\mathrm{video}} = S_{\mathrm{phys}} M_{\mathrm{valid}} M_{\mathrm{measure}}, \tag{8}$$

$$R_{\mathrm{discard}} = 1 - N_{\mathrm{eff}}/N_{\mathrm{total}}. \tag{9}$$

For model m and metric k, the effective-video score $\bar{S}^{\mathrm{eff}}_{mk}$ averages S_{video} over the N_{eff} effective videos (end-to-end means over N_{total} are in Appendix E.1).

5 Experiments

5.1 Experimental Setup and Data Sources

We evaluate six representative T2V models, comprising four open-source models (CogVideoX-5B [4], HunyuanVideo-1.5 [5, 6], Wan2.1-1.3B [7, 33], and Wan2.2-A14B [34]) and two commercial closed-source models (Kling v1.6 [64] and Jimeng v3.0 [65]). For each model, we analyze 500 generated videos, with each video corresponding to one English prompt, yielding a total of 3,000 videos (500 × 6). No manual filtering is applied before scoring. Table E.1 identifies the model variants and native video specifications. Evaluation uses native-FPS timestamps without temporal subsampling, frame-rate conversion, spatial resizing, or truncation. The automated YOLOv8n+SAM2 tracking and physical evaluation pipeline ran on an NVIDIA RTX 4090 GPU with 24 GB VRAM, requiring approximately 4–6 hours per model's 500-video set.

5.2 Discard Rate

Discard analysis contextualizes effective-video scores by showing how often a model produces videos that are measurable under both hard gates.

Figure 3 should be read as a measurability analysis rather than a physical-accuracy ranking. Across all 3,000 videos, 1,047 (34.9%) are effective and 65.1% are discarded, leaving 56 nonempty model–metric cells (including 13 exploratory cells with $1 \leq N_{\mathrm{eff}} < 10$) and 4 empty cells out of 60. All evaluated systems discard many samples: Wan2.1-1.3B is highest at 87.2%, followed by CogVideoX-5B at 70.6%, Kling v1.6 at 64.8%, HunyuanVideo-1.5 at 61.2%, Wan2.2-A14B at 59.6%, and Jimeng v3.0 at 47.2%. Wan2.1-1.3B and CogVideoX-5B are dominated by failures of both gates,

Table 3: Effective-video means (0–100) with N_{eff} in parentheses; every cell starts with 50 videos. †: exploratory ($N_{\text{eff}} < 10$); –: no effective videos ($N_{\text{eff}} = 0$), not a zero score.

Model	Vel.	Acc.	Grav.	Fric.	Rest.
Open-source models					
HunyuanVideo-1.5 [5, 6]	8.41 (9)†	6.52 (26)	1.08 (15)	4.74 (16)	5.26 (30)
Wan2.1-1.3B [33]	7.58 (1)†	**18.67** (6)†	1.58 (10)	– (0)†	6.47 (12)
Wan2.2-A14B [34]	9.97 (11)	14.02 (23)	**3.39** (9)†	3.69 (13)	**20.10** (27)
CogVideoX-5B [4]	0.43 (27)	0.62 (18)	0.81 (30)	4.20 (1)†	1.10 (15)
Closed-source models					
Kling v1.6 [64]	5.42 (11)	9.08 (18)	2.43 (16)	4.01 (24)	11.56 (25)
Jimeng v3.0 [65]	**15.54** (42)	12.72 (35)	2.97 (28)	**5.35** (28)	8.03 (23)
Model	Dens.	Visc.	Spring	Energy	Mom.
Open-source models					
HunyuanVideo-1.5 [5, 6]	**70.46** (26)	< 0.01 (17)	11.12 (15)	3.66 (39)	0.11 (1)†
Wan2.1-1.3B [33]	< 0.01 (2)†	< 0.01 (6)†	7.31 (12)	4.13 (15)	– (0)†
Wan2.2-A14B [34]	54.86 (30)	< 0.01 (16)	9.45 (25)	**5.58** (46)	0.51 (2)†
CogVideoX-5B [4]	0.37 (11)	– (0)†	< 0.01 (25)	0.53 (20)	– (0)†
Closed-source models					
Kling v1.6 [64]	55.34 (16)	< 0.01 (6)†	8.61 (29)	4.84 (27)	**5.76** (4)†
Jimeng v3.0 [65]	70.00 (19)	< 0.01 (6)†	**11.58** (32)	4.40 (46)	1.38 (5)†

HunyuanVideo-1.5 and Wan2.2-A14B remain near the middle of the range, and Jimeng v3.0 yields the highest measurable share. These high discard rates indicate that current T2V models still struggle to reliably generate videos that satisfy the controlled conditions needed for quantitative physical evaluation.

5.3 Effective-Video Scores

Table 3 reports per-metric effective-video (conditional) means alongside effective sample counts (N_{eff}). Because effective-video scores reflect only the subset of videos passing both validity gates, they should not be viewed in isolation as an absolute benchmark ranking; instead, model performance must be interpreted holistically by combining these scores with the discard rates in Figure 3 and the effective sample sizes in each cell.

5.4 Metric Profiles

Figure 4 visualizes the per-metric effective-video scores to compare both model-level trends and metric-level difficulty. Overall, the results remain weak and uneven across metrics, and no model provides reliable quantitative control across the full metric suite. Several models nevertheless obtain relatively higher conditional scores on density: HunyuanVideo-1.5 reaches 70.46, Jimeng v3.0 reaches 70.00, and Kling v1.6 and Wan2.2-A14B also exceed 50. A plausible explanation is that this pattern may stem from a coarse qualitative physical prior rather than precise density reasoning: in many density prompts, a video may receive a comparatively high score simply by capturing the coarse rule that an object denser than the surrounding liquid should sink when the object–fluid density contrast is moderate, which can yield a trajectory closer to the target without necessarily implying controllable numerical density representation. However, such qualitative behavior does not transfer to the remaining metrics (all at most 20.10), where the evaluator requires more specific acceleration, event-timing, conservation, or terminal-motion evidence.

The leading model changes substantially across metrics: Jimeng v3.0 leads velocity, friction, and spring; Wan2.2-A14B leads gravity ($N_{\text{eff}} = 9$), restitution, and energy; HunyuanVideo-1.5 leads density; Kling v1.6 leads momentum ($N_{\text{eff}} = 4$); and Wan2.1-1.3B holds the highest conditional mean in acceleration ($N_{\text{eff}} = 6$, whereas Wan2.2-A14B and Jimeng v3.0 lead among cells with $N_{\text{eff}} \geq 10$), with apparent outliers such as Wan2.1-1.3B on acceleration or CogVideoX-5B on friction ($N_{\text{eff}} = 1$) occurring in sparse exploratory cells ($N_{\text{eff}} < 10$). Viscosity stays below 0.01 for all models, momentum suffers from severe sample attrition ($N_{\text{eff}} \leq 5$), and energy remains low even for the best model. These fragmented profiles indicate that current T2V models can capture some

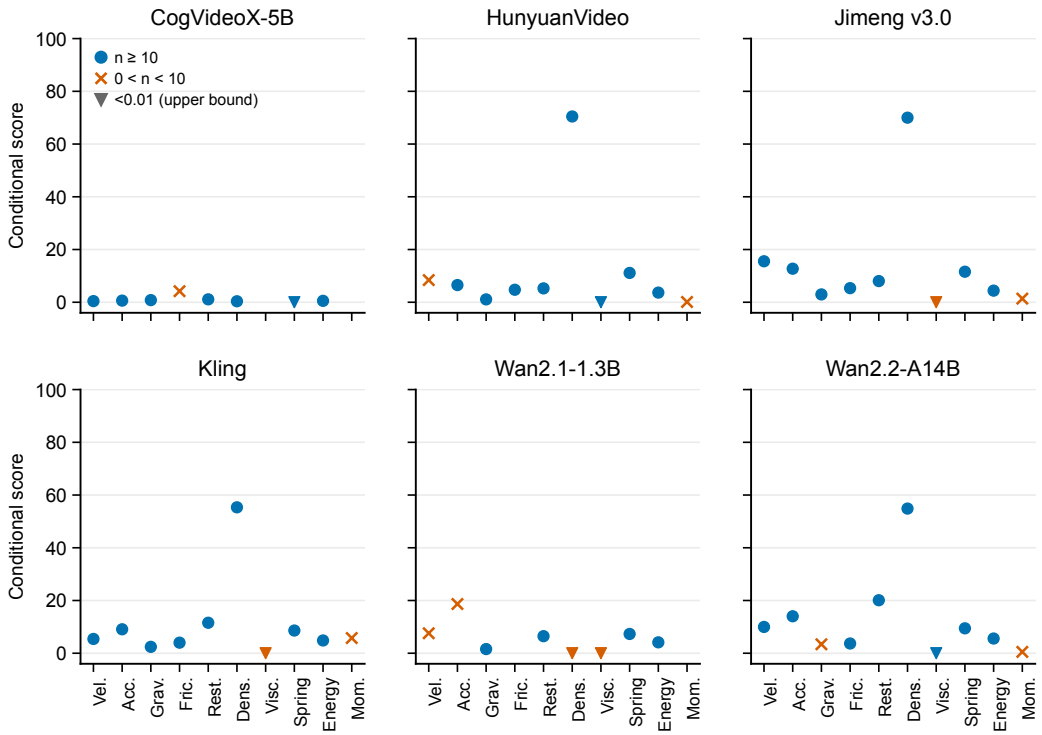


Figure 4: Conditional score profiles from Table 3. Orange markers denote $0 < N_{\text{eff}} < 10$; empty cells are omitted. Downward triangles show reported values below 0.01 at their upper reporting bound, not exact zeros.

qualitative physical cues, but do not yet generate quantitatively reliable trajectories across diverse physical laws.

Under identical video specifications, the newer and larger Wan2.2-A14B variant also achieves a lower discard rate (59.6% versus 87.2%) and higher conditional scores across most metrics than its predecessor Wan2.1-1.3B [33, 34] (Appendix E). Sensitivity analyses on alternative mask-aggregation policies and shared R^2 validity thresholds are reported in Appendix F.

5.5 Blind Human Evaluation

Five physics-trained raters independently scored all 1,047 effective videos, blinded to generator identities and automatic scores, and were compensated at approximately RMB 60 per hour. For each video, raters were shown the prompt-specified physical scenario and instructed to assess whether the observed object motion follows the requested physical law and target dynamics. The five-level scale assigns 5 to fully consistent motion and plausible quantitative dynamics; 4 to minor trajectory or timing deviations; 3 to partially consistent qualitative motion with noticeable physical inaccuracies; 2 to major physical implausibility; and 1 to a severe violation of the physical law.

Each entry in Table 4 averages the five raters' scores over the N_{eff} effective videos in that model–metric cell. Empty cells are marked "–". The conditional macro averages the K_+ nonempty metric means, with $K_+ = 8$ for CogVideoX-5B and Wan2.1-1.3B, and $K_+ = 10$ otherwise.

Across the 56 nonempty model–metric cells, human conditional means correlate strongly with the automatic effective-video means in Table 3 (Spearman $\rho = 0.80$, $p < 10^{-12}$). Human conditional macros broadly follow the automatic ordering—matching both the model with the highest observed conditional macro score (Jimeng v3.0) and the two lowest-scoring models (Wan2.1-1.3B and CogVideoX-5B)—except that Kling v1.6 ranks second in human ratings (2.93) while trailing Wan2.2-A14B and HunyuanVideo-1.5 in automatic conditional macro (10.71 versus 12.16 and 11.14). For metrics such as viscosity and gravity, human ratings remain moderate despite near-zero or very low automatic scores. This divergence reflects an inherent limitation of human perception: raters can

Table 4: Blind human-evaluation effective-video means (1–5 scale; higher is better). †: exploratory ($0 < N_{\text{eff}} < 10$); –: no effective videos ($N_{\text{eff}} = 0$). Macro averages nonempty metric means.

Model	Vel.	Acc.	Grav.	Fric.	Rest.	Dens.	Visc.	Spring	Energy	Mom.	Macro
CogVideoX-5B	2.1	2.2	2.0	2.0†	2.6	3.2	–	1.6	1.9	–	**2.20**
HunyuanVideo-1.5	2.5†	3.2	2.3	2.7	3.0	3.8	1.9	3.3	3.0	1.3†	**2.70**
Jimeng v3.0	3.5	3.5	2.5	3.2	3.3	4.1	1.8†	3.5	3.4	1.5†	**3.03**
Kling v1.6	2.9	3.0	2.4	2.9	3.2	3.7	1.8†	3.2	3.1	3.1†	**2.93**
Wan2.1-1.3B	2.3†	3.2†	2.1	–	3.1	3.3†	1.6†	2.7	2.5	–	**2.60**
Wan2.2-A14B	2.8	3.3	2.4†	2.7	3.4	3.8	2.0	3.3	3.2	1.5†	**2.84**

tell whether a motion looks qualitatively natural, but cannot accurately measure quantitative physical parameters by eye—further underscoring the necessity of strict automated evaluation.

6 Limitations and Future Work

PhysEval currently focuses on short-horizon, monocular videos under controlled benchmark conditions, enabling quantitative evaluation of T2V outputs by assuming near-planar motion parallel to the image plane and adopting simplified physical models (e.g., sliding deceleration, terminal velocity approximations, and detectable event windows). While this formulation makes physical estimation from 2D trajectories and calibration references tractable, it excludes deformable objects, complex fluids, severe occlusions, multi-view consistency, and long-horizon interactions. Moreover, generated videos frequently exhibit depth drift, scale inconsistency, camera motion, or perspective distortion that violate the planar assumption and yield physically misleading 2D trajectories—a 3D ambiguity that tracking-quality gates and mask-reliability scores only partially mitigate.

From an empirical and statistical standpoint, our validation has not yet verified the recovery of absolute physical parameters in SI units in real-world settings, nor has it measured the stochastic variance across repeated generations under fixed prompts. Furthermore, non-random sample retention from validity-gate filtering introduces selection bias and leaves certain model–metric cells with very small effective sample sizes. Consequently, PhysEval scores serve as metric-specific diagnostic indicators under the specified evaluation protocol rather than a definitive ranking of models' overall physical capabilities; while they capture broad, task-dependent performance gaps when read alongside effective counts and discard rates, small score differences and comparisons in sparse-sample regimes should be interpreted cautiously.

Future work should integrate 3D reconstruction or multi-view estimation and support richer dynamical regimes while ensuring every metric remains anchored to a verifiable physical quantity.

7 Conclusion

We presented PhysEval, a benchmark framework designed to quantitatively evaluate whether the motion dynamics in T2V generations adhere to target physical values. For each video, PhysEval produces an auditable evaluation record detailing the target and measured values, relative error, tracking-quality and physics-status diagnostics, mask-reliability scores, and discard diagnostics. Our evaluation across six video generation models reveals substantial limitations in physical adherence: no model reliably satisfies the prescribed physical constraints, a large proportion of samples are discarded by the validity gates, and even the effective videos perform poorly across most metrics. These findings underscore that current T2V models still face a vast gap between generating visually plausible motion and supporting reliable quantitative measurement of specific physical laws. We hope PhysEval serves as a valuable diagnostic tool to drive future advances in the physical controllability, measurability, and consistency of T2V generation.

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A Target Values and Frequencies

The targets form a fixed grid rather than a continuous random sample. Table A.1 summarizes the target values and prompt frequencies in the benchmark metadata; parentheses indicate prompt frequencies and each row sums to 50. Energy uses a ratio target of one. Momentum uses a relative conservation-error target of zero, rather than a target of zero absolute momentum.

Table A.1: Verified target values and prompt frequencies from the benchmark metadata.

Metric	Unit	Value (number of prompts)
Vel.	m/s	0.08 (6), 0.1 (3), 0.12 (3), 0.15 (12), 0.18 (8), 0.2 (7), 0.22 (3), 0.25 (2), 0.28 (3), 0.3 (3)
Acc.	m/s^2	-0.15 (2), -0.12 (2), -0.1 (1), -0.08 (2), -0.06 (1), -0.04 (4), 0.04 (4), 0.05 (3), 0.06 (14), 0.08 (5), 0.1 (6), 0.12 (4), 0.15 (2)
Grav.	m/s^2	9.8 (50)
Fric.	none	0.08 (12), 0.15 (13), 0.22 (13), 0.3 (12)
Rest.	none	0.35 (12), 0.55 (13), 0.75 (13), 0.85 (12)
Dens.	kg/m^3	850 (16), 1,100 (17), 1,400 (17)
Visc.	Pa·s	0.2 (12), 0.4 (13), 0.6 (13), 1 (12)
Spring	N/m	3 (6), 4 (6), 5 (6), 6 (7), 8 (6), 10 (6), 12 (7), 15 (6)
Energy	ratio	1 (50)
Mom.	relative error	0 (50)

B Physical Assumptions and Illustrated Estimators

B.1 Shared assumptions, notation, and calibration

The scene frames from Figure 2 are representative keyframes extracted from generated videos to illustrate each physical task, followed by the corresponding estimator and its applicability conditions. Exact event and period rules are collected in Appendix C.2.

The estimators use inspectable inverse models. They assume a fixed camera, approximately planar motion, a known or inferable time base, and a stable object-center track. Metric estimates additionally require a reference dimension applicable to the visible plane of motion.

Let (u_i, v_i) denote the tracked image center at timestamp t_i in frame i, where the vertical image coordinate increases downward. Each task projects (u_i, v_i) onto a 1D pixel coordinate q_i (along the horizontal or vertical axis, or the first principal component of the centered trajectory). With reference length L in calibration metadata K_i and observed pixel extent ℓ, the spatial scale α and 1D metric coordinate s_i are defined as:

$$\alpha = \frac{L}{\ell}, \qquad s_i = \alpha q_i. \tag{B.1}$$

In our evaluation pipeline, diameter calibration uses the median positive equivalent mask diameter, with a bounding-box fallback; side-length calibration uses the mean of median positive box width and height, with a diameter fallback.

For pixel-domain trajectory fitting across frames i, the linear slope and intercept $(\hat{v}_{\mathrm{px}}, \hat{c})$, quadratic coefficients $(\hat{a}_{2,\mathrm{px}}, \hat{b}, \hat{c})$, and coefficient of determination R^2 (with fitted value $\hat{q}_i$ and sample mean $\bar{q}$) are defined as:

$$(\hat{v}_{\mathrm{px}}, \hat{c}) = \arg\min_{v,c} \sum_i (q_i - vt_i - c)^2, \tag{B.2}$$

$$(\hat{a}_{2,\mathrm{px}}, \hat{b}, \hat{c}) = \arg\min_{a_2,b,c} \sum_i (q_i - a_2 t_i^2 - bt_i - c)^2, \tag{B.3}$$

$$R^2 = 1 - \frac{\sum_i (q_i - \hat{q}_i)^2}{\sum_i (q_i - \bar{q})^2}. \tag{B.4}$$

Calibrated implementations fit trajectories and assess regression validity in the pixel domain, then convert the fitted coefficients to metric units, yielding $\hat{v} = \alpha \hat{v}_{\mathrm{px}}$ and $\hat{a} = 2\alpha \hat{a}_{2,\mathrm{px}}$. Nonzero constant

scaling preserves regression R^2, although scale-dependent prerequisites can still affect estimator eligibility. For a constant trajectory, the implementation returns $R^2 = 1$ when the residual sum of squares is at most 10^{-12} and zero otherwise; the separate Tracking Quality Gate remains necessary.

B.2 Velocity



$$q(t) \approx v_{\text{px}} t + c, \qquad \hat{y}_{\text{vel}} = \alpha |\hat{v}_{\text{px}}|.$$

Figure B.1: Fit 1D trajectory $q(t)$ along the motion axis with pixel slope v_{px} and intercept c; scaling $|\hat{v}_{\text{px}}|$ by α gives speed $\hat{y}_{\text{vel}}$. Requested target: 0.28 m/s (example record `V018`).

Assumption and validity. The tracked center represents approximately uniform motion. At least four records and a metric scale are required; linear-fit R^2 follows the shared 0.65/0.85 mapping in Appendix C.1.

B.3 Acceleration



$$q(t) \approx a_{2,\text{px}} t^2 + bt + c, \qquad \hat{y}_{\text{acc}} = 2\alpha \hat{a}_{2,\text{px}}.$$

Figure B.2: Fit $q(t)$ with quadratic coefficient $a_{2,\text{px}}$, linear term b, and intercept c; scaling $2\hat{a}_{2,\text{px}}$ by α gives acceleration $\hat{y}_{\text{acc}}$ (using magnitude for nonnegative targets and retaining sign for negative targets). Requested target: 0.06 m/s^2 (example record `A029`).

Assumption and validity. Acceleration is approximately constant over the fitted interval. At least five records and a metric scale are required; the quadratic fit uses the shared R^2 mapping.

B.4 Gravity

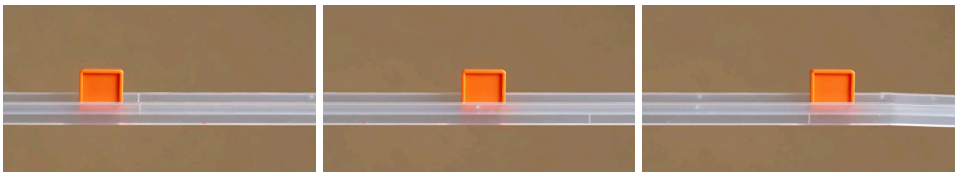


$$q_{\downarrow}(t) \approx a_{2,\text{px}} t^2 + bt + c, \qquad \hat{g} = |2\alpha \hat{a}_{2,\text{px}}|.$$

Figure B.3: Fit downward pixel displacement $q_{\downarrow}(t)$ during free fall with coefficients $a_{2,\text{px}}, b, c$ and convert to metric gravitational acceleration $\hat{g}$. Requested target: 9.8 m/s^2 (example record `G037`).

Assumption and validity. Free fall has negligible drag and no hidden support, with motion parallel to the image plane. At least five records and spatial calibration are needed for the SI task; the shared R^2 rule applies.

B.5 Friction Coefficient



$$m|a| = \mu mg, \qquad \hat{\mu} = \frac{|2\alpha\hat{a}_{2,\mathrm{px}}|}{\max(g, 10^{-12})}.$$

Figure B.4: For a slider of mass m under gravity g, infer kinetic friction $\hat{\mu}$ from calibrated deceleration $|a| = |2\alpha\hat{a}_{2,\mathrm{px}}|$. Requested target: $\mu = 0.30$ (example record `F003`).

Assumption and validity. Motion is one-dimensional sliding with approximately constant friction; rolling and active propulsion are outside this model. At least eight records, scale, and a one-dimensional score ≥ 0.75 are required. The shared fit rule applies; a strong fit without endpoint speed decrease becomes `weak_valid`.

B.6 Restitution Coefficient

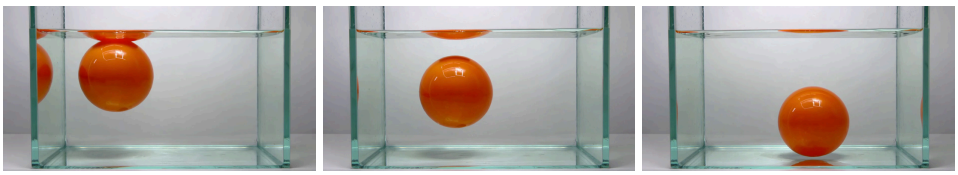


$$\hat{e} = \frac{|\hat{v}^{+}_{\mathrm{px}}|}{|\hat{v}^{-}_{\mathrm{px}}|}.$$

Figure B.5: Estimate local pixel velocities $\hat{v}^{-}_{\mathrm{px}}$ and $\hat{v}^{+}_{\mathrm{px}}$ before and after a detected horizontal reversal; their magnitude ratio defines $\hat{e}$ and cancels the common spatial scale. Requested target: $e = 0.75$ (example record `R050`).

Assumption and validity. A single impact occurs against a fixed surface; spin and tangential transfer are omitted. At least ten records and usable local windows are required. Minimum local $R^2 \geq 0.75$ is `valid`; an available lower-quality fit is `weak_valid`. Event detection uses the horizontal image axis.

B.7 Density

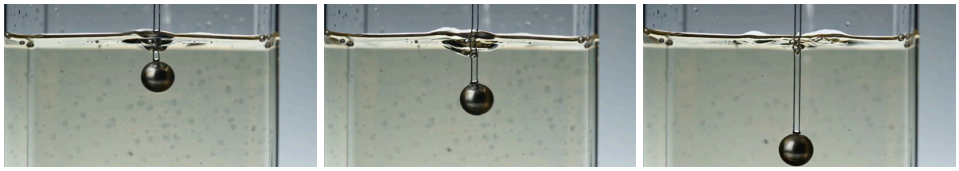


$$\rho_s V a_{\mathrm{down}} = (\rho_s - \rho_f) V g, \qquad \hat{\rho}_s = \frac{\rho_f}{1 - a_{\mathrm{down}}/\max(g, 10^{-12})}.$$

Figure B.6: Early downward acceleration $a_{\mathrm{down}} = 2\alpha\hat{a}_{2,\mathrm{px}}$ of an object (volume V, density ρ_s) balances gravity g and buoyancy in a fluid of known density ρ_f, yielding $\hat{\rho}_s$. Requested target: 1100 kg/m^3 (example record `D019`).

Assumption and validity. Drag is negligible in the early fitting interval. At least eight records, scale, fluid density, gravity, and a non-degenerate denominator are needed; the shared R^2 rule applies.

B.8 Fluid Viscosity

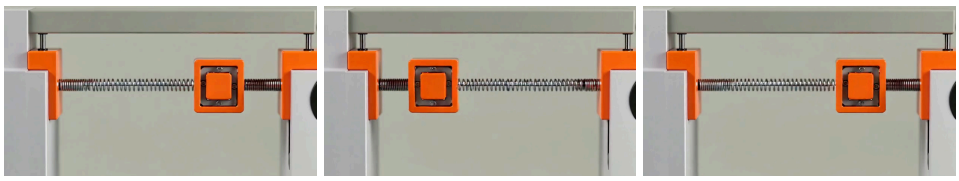


$$6\pi\eta r v_t = \frac{4}{3}\pi r^3(\rho_s - \rho_f)g, \qquad \hat{\eta} = \frac{2r^2(\rho_s - \rho_f)g}{9v_t}.$$

Figure B.7: Use a late stable-velocity interval with terminal downward speed $v_t > 0$, known sphere radius r, solid and fluid densities ρ_s, ρ_f, and gravity g to estimate dynamic viscosity $\hat{\eta}$. Requested target: 0.40 Pa·s (example record `ETA001`).

Assumption and validity. A small sphere moves in a low-Reynolds-number terminal regime. At least 12 records, scale, and positive finite viscosity are required. Accepted-window confidence ≥ 0.75 is `valid`; lower confidence is `weak_valid`.

B.9 Spring Constant

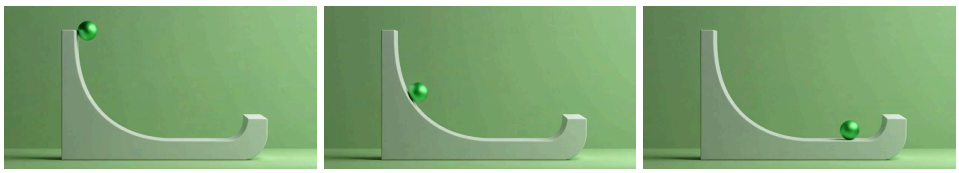


$$m\ddot{x} + kx = 0, \qquad T = 2\pi\sqrt{m/k}, \qquad \hat{k} = \frac{4\pi^2 m}{T^2}.$$

Figure B.8: For displacement x, estimate oscillation period T from peaks or autocorrelation; known mass m yields spring stiffness $\hat{k}$. Requested target: 10 N/m (example record `K047`).

Assumption and validity. Motion is approximately harmonic with a stable period. At least 12 records and positive finite mass are required; spatial scale is unnecessary. Detected-period confidence ≥ 0.20 is `valid`, otherwise `weak_valid` (see branch behavior in Appendix C.2). This heuristic, stored in `Fit_R2`, is not a regression R^2.

B.10 Mechanical Energy Conservation

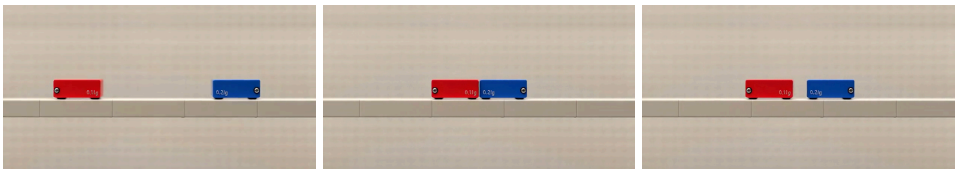


$$\Delta h = \alpha(q_{\text{low}} - q_{\text{release}}), \qquad v_{\text{low}} = \sqrt{\dot{x}^2 + \dot{y}^2}, \qquad \hat{r}_E = \frac{v_{\text{low}}^2}{2\max(g, 10^{-12})\Delta h}.$$

Figure B.9: Compare lowest-point speed v_{low} (from calibrated velocities $\dot{x}, \dot{y}$) with height drop Δh between release q_{release} and lowest point q_{low} under gravity g to estimate energy ratio $\hat{r}_E$. Requested target: energy ratio 1 (example record `E010`).

Assumption and validity. Release is from rest; the prompts specify a smooth track, so the sphere is modeled as sliding without rotation and only translational kinetic energy is used. At least ten records, scale, a later lowest point, and height drop $> 10^{-5}$ m are required. Passing prerequisites yields `valid`.

B.11 One-Dimensional Momentum Conservation



$$\hat{p}^{\pm} = m_1 \hat{v}_1^{\pm} + m_2 \hat{v}_2^{\pm}, \qquad \hat{y}_{\mathrm{mom}} = \frac{|\hat{p}^{+} - \hat{p}^{-}|}{\max(|\hat{p}^{-}|, 10^{-9})}.$$

Figure B.10: For known masses m_1, m_2, fit pre/post-collision velocities $\hat{v}_1^{\pm}, \hat{v}_2^{\pm}$ to obtain total momentum $\hat{p}^{\pm}$ and relative error $\hat{y}_{\mathrm{mom}}$ (a common scale cancels). Requested target: relative conservation error 0 (example record `P004`).

Assumption and validity. The collision is isolated and one-dimensional, with known object masses. Each track requires eight records and the event windows require at least 12 common records. Minimum local $R^2 \geq 0.70$ is `valid`; lower available fits are `weak_valid`.

C Validity Gates and Scoring

C.1 Supplementary gate implementation details

Tracking measurements. For the primary counted object, motion extent is the diagonal span of its image-space center coordinates, normalized by its median equivalent diameter. Frame coverage must use the full video frame sequence as its denominator. The standard nested tracking format records one entry per actual frame; sparse imported records require a normalization check to avoid overstating coverage.

Task-specific validity. When a measurement is available without regression R^2, the shared mapping assigns `weak_valid`; task-specific rules take precedence. For friction, an otherwise `valid` fit without endpoint speed decrease is downgraded to `weak_valid`.

Restitution uses the minimum R^2 across two pre/post-event local fits, with a `valid` threshold of 0.75. Momentum uses the minimum across four fits, with a `valid` threshold of 0.70. Available fits below these task-specific thresholds are `weak_valid`. Viscosity and spring use terminal-window and period-confidence thresholds of 0.75 and 0.20, respectively. For these two metrics, `Fit_R2` stores a detector-confidence heuristic rather than regression R^2. Energy uses prerequisite checks without an R^2 threshold.

These diagnostics assess estimator usability; agreement with the requested target is evaluated separately. Physical assumptions and estimator formulas appear in Appendix B; event, window, and period implementation rules appear in Appendix C.2.

C.2 Task-specific implementation details

The illustrated profiles retain the physical model and principal validity conditions. The following details specify the event, window, and period heuristics in our evaluation pipeline.

Gravity and friction. Ratio-only or uncalibrated gravity settings can return pixel-unit acceleration, whereas SI gravity targets use spatial calibration. A separate zero-gravity rule accepts measured acceleration below 0.2 when the reference gravity is zero (outside the benchmark grid, whose gravity targets are all 9.8 m/s^2). Friction checks whether the endpoint finite-difference speed magnitude decreases relative to the start of the trajectory.

Restitution event and windows. Horizontal finite-difference velocity is smoothed with a five-record moving average. Candidate sign flips require nonzero velocities on both sides (magnitude above 10^{-6}). The strongest candidate by the sum of neighboring speed magnitudes is selected. With $w = \max(4, \min(10, \lfloor n/6 \rfloor))$, local slope fits use clipped half-open index windows $[i - w, i - 1)$

and $[i+1, i+w+2)$. Each fit requires at least three points. The pre-event speed must exceed 10^{-9} px/s; missing events or unusable windows fail estimation.

Density early window. The first $\max(6, \min(n, \lfloor n/3 \rfloor))$ records are fitted. A buoyancy-denominator magnitude at most 10^{-9} fails estimation.

Viscosity terminal window. Finite-difference velocity is smoothed with five records. Set $h = \max(5, \lfloor n/5 \rfloor)$ and search late windows $[j, j+h)$ for $j = \max(1, n-2h), \ldots, n-h$. For each window, compute $r_j = \mathrm{std}(v)/|\,\mathrm{mean}(v)|$. The selected window minimizes $r_j + 0.02(n-j)$. Windows with near-zero mean are skipped; detection fails if the selected $r_j > 0.35$. Confidence is $1 - \min(r_j, 1)$. An accepted estimate additionally needs $v_t > 10^{-12}$ and positive finite viscosity. An upward trajectory is not converted to downward motion by taking its magnitude.

Spring peak detection. At least 12 records and finite positive mass are required. The mean-centered projected displacement is passed to a peak detector with minimum separation $\max(2, \lfloor 0.15n \rfloor)$ and prominence 0.15 times its standard deviation. With at least two peaks, T is the median timestamp interval between peaks and confidence is N_{peaks}/n.

Spring autocorrelation fallback. If peak detection is unavailable, fails, or yields fewer than two peaks, autocorrelation is used. With median timestamp spacing $\Delta t > 0$, the minimum lag is $\max(2, \lfloor 0.25/\Delta t \rfloor)$. Select the first positive local maximum beyond that lag, or the largest remaining correlation if no such maximum exists. Then $T = \ell \Delta t$; after zeroing lag zero, confidence is the selected correlation divided by the largest absolute correlation, with a 10^{-12} denominator floor. Fallback confidence below 0.15 fails detection. A detected period is `valid` at confidence ≥ 0.20 and otherwise `weak_valid`.

Spring confidence interpretation. Under the $\lfloor 0.15n \rfloor$ separation rule, standard-length peak detections have $N_{\mathrm{peaks}}/n < 0.20$ (e.g., $4/120 \approx 0.033$) and are weighted as `weak_valid`, whereas the 0.20 threshold primarily filters autocorrelation fallback estimates (`Fit_R2` stores this branch-dependent heuristic rather than regression R^2).

Energy event selection. At least ten records and scale are required. The first tracked position is the operational release position; the maximum downward image position selects the lowest point. Velocity is computed by calibrated finite differences. The lowest-point index must exceed one and the height drop must exceed 10^{-5} m. Passing these prerequisites yields `valid`, with no `weak_valid` status.

Momentum event and windows. Tracks are aligned in time and the closest image-space approach selects the event. The first track defines a common PCA axis for both velocity estimates. With $w = \max(4, \min(10, \lfloor n/5 \rfloor))$, require $i - w \geq 2$ and $i + w + 2 \leq n$. Local fits use $[i-w, i-1)$ and $[i+1, i+w+2)$; each needs at least three points. The boundary checks imply at least 12 common records for a usable event.

C.3 Planarity and mask reliability score

The physical estimators use 2D image trajectories and assume approximately planar motion with a fixed camera. Because generated videos may exhibit depth variation, perspective changes, or mask instability, we use $Q_{\mathrm{mask}} \in [0, 1]$ as a continuous reliability weight to downweight trajectories with scale, shape, consistency, or completeness degradation.

Ten diagnostics are evaluated over the middle 15–85% of frames (Table C.1); box width and height CV are separate components. For object o and diagnostic j, a high-is-bad value d_{oj} with good/bad thresholds g_j, b_j and exponent $\gamma = 0.35$ defines the component score q_{oj} as:

$$q_{oj} = \begin{cases} 1, & d_{oj} \le g_j, \\ 0, & d_{oj} \ge b_j, \\ 1 - \left(\dfrac{d_{oj} - g_j}{b_j - g_j} \right)^{\gamma}, & g_j < d_{oj} < b_j, \end{cases} \qquad \gamma = 0.35. \tag{C.1}$$

Low-is-bad diagnostics use the reverse direction, with $q_{oj} = 1$ above the good threshold and zero below the bad threshold. Between these thresholds, $q_{oj} = 1 - [(g_j - d_{oj})/(g_j - b_j)]^{0.35}$.

For each tracked object o with available component set $\mathcal{A}_o \neq \emptyset$, the object score u_o, video-level mask reliability Q_{mask}, and measurement multiplier M_{measure} (with tracking indicator M_{track}) are defined as:

$$u_o = \min_{j \in \mathcal{A}_o} q_{oj}, \qquad Q_{\mathrm{mask}} = \min_{o} u_o, \qquad M_{\mathrm{measure}} = M_{\mathrm{track}} Q_{\mathrm{mask}}. \tag{C.2}$$

Missing or nonfinite individual diagnostics are ignored, while an insufficient-valid-middle-frames flag contributes a zero component and a low-middle-coverage flag does likewise when no measured coverage component is available. Every scored video must contain object-level diagnostics with at least one computable component per object ($\mathcal{A}_o \neq \emptyset$); missing object diagnostics halt scoring rather than receiving a default fallback value, and quality-control flags never raise component scores. Under the default soft policy, mask reliability rescales the adjusted score without changing effective eligibility. A computable $Q_{\mathrm{mask}} = 0$ therefore gives zero contribution while the video remains effective if both the tracking-quality and physics-status gates pass.

Table C.1: Mask-reliability thresholds used by the scoring implementation. High-is-bad diagnostics receive full credit at or below the good threshold and zero at or above the bad threshold. Low-is-bad diagnostics use the reverse direction. Width and height CV are scored separately.

Diagnostic	Good	Bad	Direction
Area ratio p95/p05	2.00	4.00	High is bad
Area CV	0.25	0.70	High is bad
Box width / height CV	0.20	0.60	High is bad
Box aspect CV	0.20	0.60	High is bad
Maximum log-area jump	0.45	1.20	High is bad
Trend scale ratio	1.60	3.00	High is bad
Maximum centroid step / diameter	2.50	5.00	High is bad
Translation-aligned IoU	0.65	0.30	Low is bad
Middle-window coverage	0.70	0.50	Low is bad

D Validation Experiments

We evaluate 123 real physical-experiment videos from Morpheus [20] to examine tracking. Tracking passed in $117/123$ cases (95.1%; Wilson 95% interval 89.8–97.7%).

Within the same collection, seven target-conditioned videos initialized with public clicks had $7/7$ tracking passes. Eleven falling-ball videos matched to the observation assumptions passed in all cases. Their vertical quadratic fits had median $R^2 = 0.999919$ (range 0.999790–0.999931). These two subsets are drawn from the same 123-video collection (Table D.1).

Table D.1: Summary and interpretation of the real-video control checks.

Population	Observation	Supported interpretation
Full collection ($n = 123$)	117 tracking	Usability under the real-video control protocol
Public-click subset ($n = 7$)	7 tracking	Tracking under the specified initialization
Falling-ball subset ($n = 11$)	11 passes; high quadratic R^2	Constant-acceleration trajectory shape in image coordinates

E Extended Results

Table E.1: Evaluated model variants and native video specifications (500 videos per model). Duration equals frame count divided by FPS, including the final frame interval.

Model	Variant / version	$W \times H$	FPS	Frames	Duration (s)
CogVideoX-5B	`CogVideoX-5B`	1360×768	8	41	5.1250
HunyuanVideo-1.5	`HunyuanVideo-1.5`	1280×720	24	121	5.0417
Wan2.1-1.3B	`Wan2.1-T2V-1.3B`	832×480	15	81	5.4000
Wan2.2-A14B	`Wan2.2-T2V-A14B`	832×480	15	81	5.4000
Kling v1.6	T2V v1.6	1280×720	24	121	5.0417
Jimeng v3.0	T2V v3.0	1248×704	24	121	5.0417

Wan2.1-1.3B is a 1.3B dense model, whereas Wan2.2-A14B uses a two-expert MoE with approximately 27B total and 14B active parameters [33, 34]. The two evaluated variants share the same resolution, FPS, frame count, and duration (Table E.1).

E.1 Effective-video and end-to-end estimands

For model m and metric k, let E_{mk} denote the effective video set with count $N_{mk}^{\text{eff}} = |E_{mk}|$. Eligibility requires passing the Tracking Quality Gate, obtaining a `valid` or `weak_valid` status from the Physics Status Gate, and yielding a computable adjusted video score $S_{\text{video},i}$ (a zero mask-reliability weight or `weak_valid` weight does not remove eligibility). For a nonempty cell ($N_{mk}^{\text{eff}} > 0$), the effective-video mean $\bar{S}_{mk}^{\text{eff}}$ and end-to-end (E2E) mean $\bar{S}_{mk}^{\text{E2E}}$ over all 50 prompts are defined as:

$$\bar{S}_{mk}^{\text{eff}} = \frac{1}{N_{mk}^{\text{eff}}} \sum_{i \in E_{mk}} S_{\text{video},i}, \qquad \bar{S}_{mk}^{\text{E2E}} = \frac{N_{mk}^{\text{eff}}}{50} \bar{S}_{mk}^{\text{eff}}. \tag{E.1}$$

An empty cell has no conditional estimate and an E2E mean of zero. The conditional model macro averages only its K_+ nonempty metrics ($K_+ = 8$ for CogVideoX-5B and Wan2.1-1.3B, and $K_+ = 10$ otherwise), whereas the E2E macro averages all ten metrics. Across any evaluated cohort of N_{total} videos with N_{eff} effective videos, the discard rate R_{discard} is defined as:

$$R_{\text{discard}} = 1 - \frac{N_{\text{eff}}}{N_{\text{total}}}. \tag{E.2}$$

The saved cohort contains 1,047 effective videos out of 3,000 (34.9%). Of 60 model–metric cells, 13 have one to nine effective videos and four are empty. Conditional scores describe performance on the measurable subset, whereas E2E scores combine eligibility and scored accuracy across all 50 prompts per metric.

E.2 Relation to the main-text results

The main-text effective-video score table provides all per-metric conditional means, effective counts, and sparse-cell markers, while the tables below report E2E scores and model-level macro summaries.

E.3 End-to-end results

Table E.2: Per-metric E2E means on the 0–100 scale. Each cell averages all 50 videos, with non-effective outcomes assigned zero. Values are reconstructed from the displayed conditional means and counts in Table 3; < 0.01 denotes an upper bound.

Model	Vel.	Acc.	Grav.	Fric.	Rest.	Dens.	Visc.	Spring	Energy	Mom.
CogVideoX-5B	0.23	0.22	0.49	0.08	0.33	0.08	0.00	< 0.01	0.21	0.00
HunyuanVideo-1.5	1.51	3.39	0.32	1.52	3.16	36.64	< 0.01	3.34	2.85	< 0.01
Jimeng v3.0	13.05	8.90	1.66	3.00	3.69	26.60	< 0.01	7.41	4.05	0.14
Kling v1.6	1.19	3.27	0.78	1.92	5.78	17.71	< 0.01	4.99	2.61	0.46
Wan2.1-1.3B	0.15	2.24	0.32	0.00	1.55	< 0.01	< 0.01	1.75	1.24	0.00
Wan2.2-A14B	2.19	6.45	0.61	0.96	10.85	32.92	< 0.01	4.73	5.13	0.02

E.4 Macro scores and effective counts

Table E.3: Approximate macro scores reconstructed from the displayed conditional means (0–100), effective counts, and discard rates. Conditional macro averages the K_+ nonempty metrics; E2E averages all ten metrics with non-effective outcomes assigned zero. The displayed range reflects rounding bounds from < 0.01 entries.

Model	N_{eff}	K_+	Discard (%)	Conditional macro	E2E macro
CogVideoX-5B	147/500	8	70.6	1.01	0.16–0.17
HunyuanVideo-1.5	194/500	10	61.2	11.14	5.27
Jimeng v3.0	264/500	10	47.2	13.20	6.85
Kling v1.6	176/500	10	64.8	10.71	3.87
Wan2.1-1.3B	64/500	8	87.2	5.72	0.73
Wan2.2-A14B	202/500	10	59.6	12.16	6.39

E.5 Sparse cells and interpretation

Because each prompt–model pair is evaluated on a single generation over a fixed target grid, these scores reflect performance on the retained cohort rather than stochastic generation variance (see Section 6 for a broader discussion of limitations).

F Scoring Configuration Sensitivity

We examine the minimum-based mask policy and the shared R^2 validity thresholds using the completed sensitivity experiments described below. Mask reliability and score aggregation follow the main-text definitions and Appendix C.3. These comparisons hold generated videos and measured physical quantities fixed.

F.1 Mask-reliability distribution

Across the 1,047 effective videos under the continuous-min policy, video-level mask reliability Q_{mask} is concentrated at the extremes: 527 videos (50.33%) attain $Q_{\text{mask}} = 1$, 315 (30.09%) receive $Q_{\text{mask}} = 0$, and the remaining 205 (19.58%) are distributed across $(0, 0.25]$ (95; 9.07%), $(0.25, 0.50]$ (78; 7.45%), $(0.50, 0.75]$ (30; 2.87%), and $(0.75, 1)$ (2; 0.19%). The 315 videos with $Q_{\text{mask}} = 0$ remain effective and contribute zero adjusted scores, whereas videos failing either hard gate are excluded from this distribution and enter E2E averaging with zero scores.

F.2 Alternative mask policies

Table F.1 compares model rankings across four mask-aggregation policies. Geometric aggregation preserves the exact ordering of our default continuous-min policy, whereas arithmetic aggregation and hard-fail induce only localized swaps between the bottom pair (Wan2.1-1.3B and CogVideoX-5B) and the highest-scoring pair (Jimeng v3.0 and Wan2.2-A14B), respectively. Although varying the mask policy introduces minor shifts, our continuous-min strategy enforces a stricter quality standard by penalizing the weakest diagnostic component, thereby ensuring more rigorous filtering of unstable masks. Moreover, because these adjustments are strictly confined to adjacent pairs, the observed macro-score ordering across models remains highly stable.

Table F.1: Model-order sensitivity to mask policy.

Mask policy	Reported model order
Continuous-min	Jimeng v3.0 > Wan2.2-A14B > HunyuanVideo-1.5 > Kling v1.6 > Wan2.1-1.3B > CogVideoX-5B
Continuous-geometric	Same as continuous-min
Continuous-arithmetic	Only Wan2.1-1.3B and CogVideoX-5B exchange positions
Hard-fail	Only Jimeng v3.0 and Wan2.2-A14B exchange positions

F.3 Shared validity-threshold sensitivity

We tested the 3×3 grid of weak R^2 cutoffs $\{0.60, 0.65, 0.70\}$ and valid cutoffs $\{0.80, 0.85, 0.90\}$. The study covers velocity, acceleration, gravity, and density (1,200 videos), with other settings fixed. Friction also uses the shared R^2 rule but is excluded from this scan. The baseline $0.65/0.85$ thresholds reproduce all 1,200 saved statuses. Spring retains its separate period-confidence rule.

Table F.2: Shared R^2 threshold sensitivity. Counts cover the 1,200 videos in the four scanned metrics (friction is not scanned). N_{phys} counts valid plus weak-valid estimates; N_{eff} additionally requires tracking pass and a computable score. ΔN_{eff} is relative to the baseline 439; ρ_{E2E} is the Spearman rank correlation between the six-model E2E macro rankings (all ten metrics) under each setting and under the baseline; Top model is the model ranked first by E2E macro. The baseline row is bold.

R^2_{weak}	R^2_{valid}	$N_{\text{valid}}/N_{\text{weak}}$	N_{phys}	N_{eff}	ΔN_{eff}	ρ_{E2E}	Top model
.60	.80	642 / 124	766	454	$+3.4\%$	1.000	Jimeng v3.0
.60	.85	601 / 165	766	454	$+3.4\%$	1.000	Jimeng v3.0
.60	.90	543 / 223	766	454	$+3.4\%$	1.000	Jimeng v3.0
.65	.80	642 / 105	747	439	0.0%	.943	Jimeng v3.0
.65	**.85**	**601 / 146**	**747**	**439**	**0.0%**	**1.000**	**Jimeng v3.0**
.65	.90	543 / 204	747	439	0.0%	1.000	Jimeng v3.0
.70	.80	642 / 82	724	429	-2.3%	.943	Jimeng v3.0
.70	.85	601 / 123	724	429	-2.3%	.943	Jimeng v3.0
.70	.90	543 / 181	724	429	-2.3%	.943	Jimeng v3.0

Across the nine settings, the effective count changes by -2.3% to $+3.4\%$, E2E rank correlations range from 0.943 to 1, and Jimeng v3.0 remains first. These results indicate limited ranking changes within the tested grid for the four scanned metrics; the thresholds remain operational rather than ground-truth-calibrated.

G Qualitative Gate-Failure Diagnostics

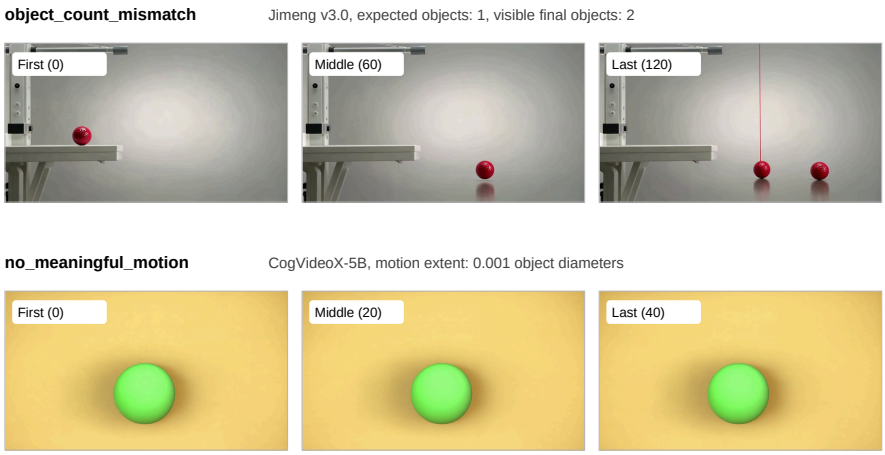


Figure G.1: Representative tracking-quality gate failures shown via first, middle, and last video frames. **Top (`object_count_mismatch`):** Jimeng v3.0 is prompted to generate a single falling red golf ball, but a second ball appears in the final frames (expected: 1, detected: 2). **Bottom (`no_meaningful_motion`):** CogVideoX-5B is prompted to generate a green basketball decelerating at -0.06 m/s^2, but the ball remains nearly stationary throughout the sequence (motion extent: $0.001 < 0.75$ object diameters).

Figure G.1 illustrates two representative failure modes rejected by the Tracking Quality Gate: an object-count violation (`object_count_mismatch`) and insufficient displacement (`no_`

meaningful_motion). While the sampled keyframes visually highlight these generation anomalies, actual gate rejection in our pipeline is determined automatically from full-sequence trajectory diagnostics, including persistent object counts, frame coverage, and normalized motion extent.

H Existing Asset Licenses and Terms

Table H.1 identifies the six evaluated video generators, the tracking components used in the main and comparison pipelines, and the real-video validation dataset. Citations identify the original papers or official model documentation; linked license names lead to the providers' published terms. Model weights, repository code, and hosted services can have different licensing arrangements.

Table H.1: Main external model, software, and dataset assets used by PhysEval. Licenses refer to the specified asset or release; proprietary services are governed by service agreements.

Asset	Role in this work	Citation or owner	License or terms
HunyuanVideo-1.5	Evaluated T2V model	Tencent Hunyuan [5, 6]	Tencent Hunyuan Community License Agreement.
Wan2.1-1.3B	Evaluated T2V model	Wan Team [7, 33]	Apache License 2.0.
Wan2.2-A14B	Evaluated T2V model	Wan Team [34]	Apache License 2.0.
CogVideoX-5B	Evaluated T2V model	THUDM / Zhipu AI [4]	CogVideoX model license for 5B weights; Apache License 2.0 for repository code and the 2B release.
Kling v1.6	Evaluated T2V service	Kuaishou Technology [64]	Proprietary service; applicable Kling AI service/API terms.
Jimeng v3.0	Evaluated T2V service	Jimeng AI [65]	Proprietary service; user-service and paid-service agreements, where applicable.
YOLOv8n	Object proposal initialization	Ultralytics [60]	AGPL-3.0 or Ultralytics Enterprise License, according to the applicable license arrangement.
SAM2	Video mask propagation	Meta AI [61]	Apache License 2.0 for SAM2 code and checkpoints.
Faster R-CNN	Detector comparison	Ren et al. [62]; TorchVision	BSD-3-Clause for TorchVision code; pretrained weights follow TorchVision's applicable weight terms.
TAM	Tracking comparison	Yang et al. [63]	MIT for Track-Anything code; upstream components retain Apache-2.0 for SAM1 and MIT for XMem.
Morpheus	Real-video validation dataset	Zhang et al. [20]	MIT, as declared in the official real-world dataset card.

Each external asset remains governed by its respective open-source license or commercial service agreement as listed in Table H.1.

I Broader Impacts

The broader goal of PhysEval is to make the physical reliability of T2V models more transparent for education, simulation prototyping, robotics-data generation, and scientific visualization. Separating

measurement eligibility from numerical target agreement can help users distinguish visually appealing motion from quantitatively supported behavior.

Benchmark scores should not be treated as certification for safety-critical deployment. Generated videos may look plausible while violating physical laws, and the measurable subset does not represent every generated outcome. Overstating benchmark performance could encourage inappropriate use of T2V outputs as reliable simulations or training data without independent validation. Such applications require task-specific checks against real measurements and attention to unmeasurable or failed outcomes.

NeurIPS Paper Checklist

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   Answer: [Yes]

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   Answer: [Yes]

   Justification: Section 6 discusses monocular planar assumptions, simplified physical models, 3D projection ambiguities, real-world SI validation and multi-seed variance, selection bias from validity-gate filtering, and cautions against overinterpreting sparse-sample scores.

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